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# ***PRINCIPLES OF ANIMAL ECOLOGY (ZOO507)***

## ***CURRENT MIDTERM PAPER FALL 2023(12-DEC)***

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### **SHORT QUESTIONS(3)**

#### **1. Natural sources of Acid Rain**

- Volcanic emissions
- Wildfires
- Lighting
- Decaying vegetation

#### **2. Coniferous Forests Climate :**

Climate is cold. Long and harsh Winters is for more than six months. Mean annual temperature is below 0°C,

#### **3. Climate of Temperate Deciduous Forests :**

These forests occur in the areas of moderate climatic conditions such as Annual rainfall is 75 to 150 cm, Winter lasts for four to six months, Temperature ranges between 10 to 20°C.

#### **4. Distribution of marine ecosystem**

Marine ecosystem covers nearly 71 % of the earth's surface with an average depth of about 4000 m. Fresh water rivers eventually empty into ocean. Salinity of open sea is 3.6% is quite constant. Sodium and chlorine-86% of the sea salt and the rest is due other elements such as sulphur, magnesium, potassium and calcium.

#### **5. Climate of marine ecosystem**

Temperatures in the ocean range from just around freezing at the pole and in the deep waters, to tropical clear waters that are as warm as a bathtub. The average temperature of all oceans is about 39°F (4°C). Heat from the sun warms only the surface of the water. Deep down, oceans everywhere are cold and dark.

#### **6. Sand Dunes:**

- ☐ Extensive areas of undulating sand dunes, absence of cultivation and extensive semi desert.
- ☐ Thal desert, Thar and Cholistan desert, Nuski and Chagai sand dunes areas.

#### **7. Ecosphere**

- largest and nearly self sufficient biological system Including all living organisms of earth interacting with the physical environment (Lithosphere-the Soil, Hydrosphere-water, Atmosphere-air whereas Biosphere- life zone of the earth)

Ecosphere = Biosphere + Lithosphere + Hydrosphere + Atmosphere)

#### **8. Define ecology and geology**

**Ecology** (from Greek word: oikos, "house", or "environment" and logos means "study of") 2. Ecology is the scientific study of the interactions between organisms and their environment

**Geology** is the scientific study of the Earth's solid materials, including rocks, minerals, and the processes that shape the Earth's structure.

#### **9. Scope Of Ecology**

- all organisms living on Earth
- all physical and chemical surroundings.
- for this reason, the field is usually divided into different levels of study including: organismal ecology, population ecology, community ecology and ecosystem ecology.

#### **10. Branches of Ecology**

- Applied ecology, (includes agroecology and conservation biology)
- Biogeochemistry
- Biogeography
- Conservation ecology

- Ecological succession
- Evolutionary ecology
- Functional ecology,
- Global ecology
- Marine ecology, and fresh water ecology
- Microbial ecology
- Paleoecology
- Restoration ecology plan

## 11. Decomposers

Feed on dead and decayed plants or animals. Make up the final trophic level in food chain. They decompose the dead and decay matter and helps in recycling the nutrients.

They are classified into two class

- 1) **Micro-Decomposers:** bacteria' fungi' protozoa
- 2) **Macro-Decomposers:** earth worm' nematodes' molluscs

## 12. Difference b/w directly and indirectly available resources.

**Directly available resources** are those that can be readily accessed and utilized by organisms without the need for any additional processing or conversion. Examples include sunlight, air, water, and food sources like fruits or nuts.

**Indirectly available resources**, on the other hand, require some form of processing or conversion before they can be used by organisms. Examples include wood that needs to be converted into fuel or building materials, minerals that need to be extracted and refined, or agricultural crops that need to be harvested and processed before they can be consumed.

## 13. How the abundance of ozone in stratosphere and troposphere determined?

Ozone abundances in the stratosphere and troposphere are determined by,

- Amount of reactive gases
- Sunlight intensity
- Temperature

## 14. Causes of global warming

- Pollution from coal, natural gas and oil
- Carbon dioxide from airplanes and vehicles
- Population increase
- Deforestation

## 15. Effect of global warming

Global warming means that the Earth is getting hotter because of human activities. When the Earth gets hotter, it causes problems like rising sea levels, really hot days, and changes in weather patterns. It can also hurt animals and plants by changing where they live and making it harder for them to survive.

## 16. What are the Salient features of Baluchistan desert scrub forest .

**Baluchistan Desert Scrub Forests:**

Usually associated with higher hills, stony plateaus. Very cold winters, no monsoon influence but occasional winter and spring showers. Typified by Northern Kalat, lower parts of Suleiman hills, most of the Balochistan valleys, Kurram valley, most of Waziristan. Heavily overgrazed.

**Fauna:** Hill fox, Balochistan race of Black bear, Hyaena, Leopard, Marbled Pole Cat, caracal Cat, Urial, Goitred Gazella, Tibetan Hare, porcupine, Migratory Hamster, Baluchistan Gerbil

## 17. Pakistan National Animal Markhor

- The markhor-national animal of Pakistan.
- Pakistan's national animal, "Markhor" symbol of bravery and sharp intelligence.
- Haven't any fear during prey and jump from the peak of the mountain to the base.
- It was one of the 72 animals featured on the WWF Conservation Coin Collection in 1976.
- Markhor is the National animal of Pakistan.
- Scientific name: *Capra falconeri falconeri*

## 18. IDA

- U.S.-based International Dark Sky Association (IDA), formed in 1988 to preserve the natural night sky

- In 2017, the IDA approved the first U.S. dark sky reserve
- December of 2018, IDA lists thirteen dark sky reserves on their site.

## 19. Radioactive contamination

• Also called radiological contamination, is the deposition of, or presence of radioactive substances on surfaces or within solids, liquids or gases (including the human body), where their presence is unintended or undesirable

## 20. Radioactivity

“Radioactivity is a phenomenon of spontaneous emission of protons (alpha particles), electrons (beta particles) and gamma rays (short wave electromagnetic waves) due to the disintegration of atomic nuclei of some elements. These cause Radioactive Pollution”.

## 21. Pollution and its three types

### ❖ Definition:

The presence in or introduction into the environment of a substance or thing that has harmful or poisonous effects **Or** the addition of any substance (solid, liquid, or gas) or any form of energy (such as heat, sound, or radioactivity) to the environment at a rate faster than it can be dispersed, diluted, decomposed, recycled, or stored in some harmless form.

### ❖ Types

Three major forms of environmental pollution > Air pollution > water pollution > land pollution > light pollution > litter > Noise pollution > Plastic pollution > Soil contamination > Radioactive contamination > Thermal pollution > Visual pollution

## 22. Point Pollutants and it's 3 examples

Point sources are single, identifiable sources.

**Examples** are,

- Smokestack of a coal-burning power or industrial plant,
- The drainpipe of a factory,
- The exhaust pipe of an auto mobile.

## 23. Noise Pollution

Noise is the unpleasant and undesirable sound which leads to discomfort in human beings.

- The intensity of sound is measured in decibels (dB).
- The faintest sound which can be heard by the Human ear is 1 Db

## 24. Effects of Air Pollution

- Smog
- Acid rain
- Eutrophication
- Global warming

## 25. Sources of land pollution/Types of land Pollution

- Mining waste
- Agricultural Waste-
- Industrial Waste
- Municipal Solid Waste

## 26. How Mining waste is generated?

It is generated by 3 main ways

**a) Mining operations-** large rocks needs to be removed to get valuable ore. Waste material kleft at site

**b) Milling operations-** through various technologies Waste material is called tailing

c) Failure of earthen dams

## 27. Lake water pollution causes

- Lakes are the rich source of water but due to following reasons, lake water gets polluted:

- Discharge of organic waste of homes and factories.
- Discharge of organic matter of sewage treatment plants.
- Dumping of garbage of villages and cities.

- Inorganic pollutants of fertilizers and pesticides used in agriculture

## 28. Surface water pollution causes

### Runoff From Paved Surfaces

➤ Urban runoffs discharged into surface water from paved surfaces such as streets and parking lots can make the water warmer. ➤ During the summer season, the pavement gets quite hot, creating warm drains that enter sewers and bodies of water.

## 29. Water pollution sources

Water pollution can come from different sources. Industrial waste discharge, agricultural runoff, sewage and wastewater, oil spills, improper disposal of chemicals and hazardous waste, stormwater runoff, mining activities, atmospheric deposition of pollutants, leakage from underground storage tanks, and marine debris and litter can all contribute to water pollution. These sources release harmful substances into water bodies, which can have negative effects on aquatic life and human health.

## 30. Difference between spring and neap tides.

Spring tides occur twice each lunar month all year long without regard to the season. Neap tides, which also occur twice a month, happen when the sun and moon are at right angles to each other. The spring tides have higher high tides and lower low tides, whereas neap tides have lower high tides and higher low tides. The difference in the range of the water level between high and low tide is much larger in a spring tide than in a low tide.

## LONG QUESTIONS(5)

### 1. Threats to snow leopards:

- Habitat degradation and fragmentation
- Reduction of natural prey due to illegal hunting
- Retribution killing for livestock loss
- War and related military activities

### 2. Community Ecology-Types of Interaction:

Interactions between same species in a community are called **intraspecific** “intra” means “same.”  
**Example:** Mating.

Interactions between different species in a community are called **interspecific interactions**— inter means "between." **Example:** One example is mutualism, where both species benefit, like the relationship between bees and flowers. Bees collect nectar from flowers while pollinating them, helping both species.

### 3. Commensalism: (+/0 interaction)

Two species have a long-term interaction that is beneficial to one and has no positive or negative effect on the other. Sucker fish, often attaches to a shark by means of its sucker which is present on the top of its head. This helps fish to get protection, a free ride as well as a meal from the left over of the shark's meal.

**Example-** grazing cattle and cattle egrets. As the cattle graze in the grass, they stir up the insects living there, allowing the cattle egret a tasty meal. The cattle egrets get a meal, but the cattle receive nothing in return from the long-necked birds, nor are they harmed by the relationship.

### 4. Amensalism: (-/0,Interaction) with 2 examples.

It describes an interaction in which the presence of one species has a negative effect on another, but the first species is unaffected.

**Example:**

- 1) a herd of elephants walking across a landscape may crush fragile plants
- 2) A tree casting a dense shadow that prevents other plants from receiving sunlight.
- 3) An antibiotic-producing fungus inhibiting the growth of nearby bacteria.

### 5. Difference b/w biotic and abiotic factors

#### ❖ Biotic Factors

Biotic factors refer to living things or organisms that directly or indirectly influence an ecosystem. They include plants, animals, fungi, bacteria, and other microorganisms. Biotic factors interact with each other and with the environment, shaping the dynamics of an ecosystem.

## ❖ Abiotic Factors

On the other hand, abiotic factors are non-living components of an ecosystem. These factors include physical and chemical elements such as temperature, sunlight, water, soil composition, air quality, and pH levels. Abiotic factors play a crucial role in determining the characteristics and availability of resources in an ecosystem.

### 6. Write three strata of tropical grassland.

**(i) Root layer**-mostly root biomass-upper 16cm, roots may penetrate 1.7m down-underground stems or rhizomes

**(ii) Ground layer**-mosses and dandelions

**(iii) Herbaceous layer**-seasonally short grasses, wild mustard, coneflower.

### 7. What is economical Growth? How it can be measured?

**Economic growth** is an increase in a nation's output of goods and services.

- It is usually **measured** by the percentage of change in a country's **gross domestic product (GDP): the annual market value of all goods and services** produced by all firms and organizations, foreign and domestic, operating within a country
- Changes in a country's economic growth per person are measured by **per capita GDP: the GDP divided by the total population at midyear**. The value of any country's currency changes when it is used in other countries

### 8. Acid Rain Types

- Wet acidic deposition
- Dry acidic deposition

**Wet Deposition:** When the wind blows the acidic chemicals in the air move to the areas where the weather is wet, the acids fall to the ground in the form of rain, sleet, fog, snow or mist. It removes acid from the atmosphere and deposits them on the earth's surface

**Dry Deposition:** If the wind blows the acidic chemicals in the air to the areas where the weather is dry, the acidic pollutants slip into dust or smoke and fall to the ground as dry particles

### 9. Health effects due to climate change

- Heart problems
- Respiratory issues
- Allergies
- Diseases spread
- Changing forest and natural areas
- Alteration in geographic distribution of vegetation type

### 10. Healths ozone . How it is depleted.

**Ozone**- outer layer of atmosphere- 20-50km above the surface of earth- stratosphere. Consist of O<sub>3</sub>  
Ozone absorb UV light (99% )

O<sub>3</sub>----- O<sub>2</sub>+O (Splitting)

O<sub>2</sub>-----2O

O<sub>2</sub>+O----- O<sub>3</sub>

**Ozone hole**- thinning of Ozone layer over the Antarctica during Southern hemisphere spring

**Chlorofluorocarbon:** 10-20 years to get into stratosphere can react with Ozone for up to 120 years

Cl+ O<sub>3</sub>-----ClO+ O<sub>2</sub>

ClO+O-----Cl+ O<sub>2</sub>

Reduction in Chlorofluorocarbon production by 50 % by the year 2000-Montreal Protocol

### Green House Effect

Light rays from sun penetrates- glass of green house – absorbed by the plant and soil-reradiate as longer wave infra red radiation (heat). Glass does not permit these rays to escape outside so heat remains within the green house

- Green house gases (CO<sub>2</sub>) are those which prevent heat to escape out from them
- Temperature increases-greenhouse effect
- Global Warming
- Melting of ice and glaciers
- Floods

## 11. What is glare and it's three types

- **Glare-**

Glare can be thought of as objectionable brightness. It can be disabling or discomforting.

- **Types**

**Blinding glare** describes effects such as that caused by staring into the Sun. It is completely blinding and leaves temporary or permanent vision deficiencies.

**Disability glare** describes effects such as being blinded by oncoming car lights, or light scattering in fog or in the eye, reducing contrast, as well as reflections from print and other dark areas that render them bright, with a significant reduction in sight capabilities.

Disability glare is when things like bright car lights or fog make it hard to see clearly. It reduces contrast and can cause reflections that make things look too bright. This makes it difficult to see properly.

**Discomfort glare** does not typically cause a dangerous situation in itself, though it is annoying and irritating at best. It can potentially cause fatigue if experienced over extended periods.

## 12. Sources of noise Pollution

- Natural sources
- Human-made sources

**Natural Sources:** volcanic eruption, the wild sea waves, a rush of rivers, thunder, roaring of animals, etc., They occur occasionally, but sometimes they are very damaging.

**Man-made Sources:** Man invented many machines for his convenience, which is now the primary cause of noise pollution

## 13. Causes of noise Pollution

- Generators, mills, huge exhaust fans
- vehicles on the roads
- Weddings, fire works, public or Political or religious gatherings involve loudspeakers
- Mining, construction of buildings
- Noise from railway and aircraft

## 14. Sources/Causes of light pollution?

- Increased energy consumption through over-illumination from artificial light sources such as street lighting, domestic lighting, and garage
- Poor planning by engineers while placing street lights and signage

## 15. Municipal Solid Waste

- Litter and Garbage
- Household waste, broken items, etc
- Commercial establishments
- 240 million metric tones/year

## 16. Solutions to Land Pollution

- Use of organic matures instead of artificial fertilizers in agriculture
- Planting of trees (Afforestation)
- Treating solid wastes such as garbage, domestic refuse, and industrial materials before dumping them in landfills
- Recovering and recycling of materials such as plastics, cloth, and glass before dumping

## 17. Groundwater Pollution

Water present below the surface of the Earth is known as groundwater. More than 90% of freshwater is Groundwater.

Groundwater pollution occurs when harmful substances, such as chemicals, heavy metals, pesticides, and bacteria, get into the water that is underground. This can happen because of different activities like industries, farming, or when waste is not disposed of properly. When these pollutants seep into the ground, they contaminate the groundwater, making it unsafe for drinking and other uses. This pollution can have negative effects on both human health and the environment. That's why it's important to take steps to prevent groundwater pollution, like practicing responsible waste management and following regulations to protect our water sources.

## 18. Effects of Radiations

### Short-range effects

- Burns
- Impaired metabolism
- Dead tissues

- Death of the organisms.
- Selectively deposit (Sr)

### **Long-range effects**

- Mutations
- Increased incidence of tumors and cancer
- Shortening of life-span
- Developmental changes.
- Selectively deposit (Sr)
- Some species of animals and plants preferentially accumulate specific radioactive materials. For example, oysters deposit (Zn), fish accumulate (Fe) marine animals

## **19. Reservoir of Phosphorus**

Phosphorus is a necessary and important constituent of the protoplasm in the living organisms ATP, ADP, AMP, NADP, Phospholipids of cell membrane 85% P found in bones and teeth's

- 1) Reservoirs of phosphorus are the rocks-formed in past geological ages
- 2) Erosion of these deposits release phosphates in the ecosystem
- 3) 3. Much of it escapes into the sea where part of it is lost to the deep sediments and some of it deposited in the shallow marine sediments
- 4) Plants take up inorganic phosphate as orthophosphate ions.
- 5) Available form of Phosphate (PO<sub>4</sub>) returns to sea
- 6) Animal consume plants-phosphorous
- 7) Death and decay (decomposition), excreta of animals, bones and teeth -Phosphorous return to nature

## **20. United Nations divided the world's countries in how many categories?**

The United Nations classifies the world's countries into four categories.

1. The developed countries (with 1.2 billion people) include the United States, Canada, Japan, Australia, New Zealand, and most countries of Europe .
2. Most are highly industrialized and have a high per capita GDP PPP
3. All other nations (with 5.5 billion people) are classified as developing countries, most of them in Africa, Asia, and Latin America
4. Some are middle-income, moderately developed countries such as China, India, Brazil, Turkey, Thailand, and Mexico. Others are low-income, least developed countries where per capita GDP PPP is steadily declining.

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